3.1.R1 (1 point possible)

Why is linear regression important to understand? Select all that apply:

- [ ] The linear model is often true
- [ ] Linear regression is very extensible and can be used to capture nonlinear effects ✔
- [x] Simple methods can outperform more complex ones if the data are noisy ✔
- [x] Understanding simpler methods sheds light on more complex ones ✔

EXPLANATION

The linear model (and every other model) is hardly ever true, but it is an important piece in many more complex methods.

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3.1.R2 (1/1 point)

Which of the following are true statements? Select all that apply:

- [x] A 95% confidence interval is a random interval that contains the true parameter 95% of the time ✔
- [ ] If I perform a linear regression and get confidence interval from 0.4 to 0.5, then there is a 95% probability that the true parameter is between 0.4 and 0.5.
- [x] The true parameter (unknown to me) is 0.5. If I sample data and construct a confidence interval, the interval will contain 0.5 95% of the time. ✔

EXPLANATION

Confidence intervals are a "frequentist" concept: the interval, and not the true parameter, is considered random. Even a Bayesian would not necessarily agree with statement 2 (it would depend on his/her prior distribution).

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